

System Architecture & Design

NV-SAD-001 · Version 1.0 · 2026-08-10 · Baseline

Field	Value
Document ID	NV-SAD-001
Product	Network Visualizer
Version	1.0
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Status	Baseline

1. Components

Component	Role
Capture	Ingest packets/connections
Flow	Aggregate rates/expiry
Process	PID mapping
Geo	IP location
Engine	Orchestrate snapshots
UI	Globe dashboard

2. Pipeline

NIC → Npcap/Connections → Flow table → Process join → Geo → Snapshot event → React store → Globe/Feed/KPIs

3. ADRs

ADR	Decision
001	Tauri over Electron
002	Full capture + fallback
003	3D globe default
004	Offline geo primary
005	Header-only
006	Snapshot IPC